ZoomRide SQL Project

Week 2 Student Guide: learn SQL with a real business story

Welcome to Week 2! I made this guide to show you exactly what to do, one step at a time. Read it slowly and follow the steps in order. You can do this.

> **Read this first. Three good news:**
> 1. You do **not** need to install anything. Everything runs in your web browser.
> 2. It is **free**, and you do **not** need to log in.
> 3. You can use a laptop or a phone. A laptop is easier.

1. What is this project about?

ZoomRide is a company that sells rides in cars and bikes, like Uber or Bolt. It works in 6 cities: **Lagos, Abuja, Port Harcourt, Nairobi, Accra and Kampala**.

The manager wants answers to 3 questions:

- Which city earns the most money?
- Which month do people ride the most?
- Which type of vehicle earns the most money?

But there is a problem. **The data is messy.** Some city names are spelled wrongly. Some trips are saved two times. Some money amounts are missing.

Your job: look at the data, clean it, find the answers with SQL, and write a short message to the manager.

2. What is SQL?

SQL (say "S-Q-L" or "sequel") is the language we use to ask questions to data. You type a short instruction, click **Run**, and the computer shows the answer.

In Week 1 you used Excel pivot tables. SQL does the same job, but you use words instead of clicks:

In Excel (Week 1)	In SQL (Week 2)
Rows = City	GROUP BY city
Values = Revenue (Sum)	SUM(fare)
Slicer (filter)	WHERE ...
Sort biggest first, show top 5	ORDER BY ... DESC LIMIT 5

We still follow our workflow: **Collect, Clean, Analyze, Decide.** This week you practise Clean, Analyze and Decide.

3. How to open it: step by step

Follow these steps in order. Do not skip any step.

1	Open your browser (Chrome, Edge or Firefox). Go to this address: **onecompiler.com/mysql**
2	Look at the top of the page. The blue button must say **MySQL**. Do not change it to anything else.
3	On the left side, find **Explorer**. Click the **New File** button (a small page with a + sign).
4	Name your file **zoomride_yourname.sql**. Example: **zoomride_amaka.sql**. Press Enter.
5	Open the file **zoomride_setup.txt** that I sent you. Select everything and copy it. (Laptop: Ctrl+A, then Ctrl+C. Phone: tap and hold, choose Select All, then Copy.)
6	Click inside your new file in OneCompiler. Paste everything. (Laptop: Ctrl+V. Phone: tap and hold, then Paste.)
7	Click the red **Run** button.
8	Check that it worked. Scroll to the bottom of the file, under the line that says **YOUR QUERIES START HERE**. Type the line below, then click Run again.

```
SHOW TABLES;
```

You should see 3 names: **customers, drivers, trips**. If you see them, you are ready!

Three important rules about the file

• Write your work **below** the line "YOUR QUERIES START HERE". Never change anything above it.

• Every time you click Run, the **whole file runs again from the top** and rebuilds the tables fresh. This is normal. So keep your queries **in order**, one after the other.

• Use **one file, one browser, one device**. Do not clear your browser history. Keep a **backup copy** of your work in Notepad or a Google Doc, so you do not lose it.

How to write a query

• Every query ends with a semicolon **;**

• A line that starts with **--** is a note to yourself. The computer ignores it. Write one before each query, like this:

```
-- Q1: Total Trips
SELECT COUNT(*) AS total_trips FROM trips;
```

4. Meet the data

A **table** is like an Excel sheet. A **row** is one record. A **column** is one type of information. We have 3 tables:

Table 1: trips (the biggest table. One row = one trip)

Column	What it means
trip_id	The number of the trip
customer_id	Which customer booked the ride
driver_id	Which driver drove
city	The city where the trip happened
trip_date	The date of the trip
distance_km	How far the trip was, in kilometres
fare	The money for the trip, in Naira (cancelled trips = 0)
status	Completed or Cancelled
payment_method	Cash, Card or Wallet

Table 2: customers (people who book rides)

customer_id, customer_name, home_city, signup_date, signup_channel

Table 3: drivers (people who drive)

driver_id, driver_name, city, **vehicle_type** (Economy, Comfort or Bike), rating, joined_date

How the tables are linked

The same number appears in two tables, like a name tag. This lets us join them together later.

```
customers.customer_id   matches   trips.customer_id
drivers.driver_id        matches   trips.driver_id
```

> **Money rule (very important)**
> Only trips with status = 'Completed' count as money. Cancelled trips earn nothing.

Words you must know

Word	Easy meaning
Dimension	A label that answers "by what?" Examples: city, month, vehicle type.
Metric	A number that answers "how much?" or "how many?" Examples: revenue, number of trips.
Duplicate	The same trip saved two times.
Missing value (NULL)	An empty box. There is no value there.
Clean data	Data where the spelling mistakes, copies and gaps have been dealt with.

5. What you must do

There are 5 parts. Do them in order. Put a note before each query with its question number and a name in the **Metric by Dimension** style from Week 1, for example: **-- Q6: Revenue by City**

Part 1: Look at the data

First, run this to see 10 rows. The * means "all columns". LIMIT 10 means "only 10 rows".

```
SELECT * FROM trips LIMIT 10;
```

Q1. How many rows are in the **trips** table?
Hint: use COUNT(*).

Q2. Show the **5 longest completed trips**. Show trip_id, city, distance_km and fare.
Hint: use WHERE to keep only Completed trips, ORDER BY distance_km DESC to put the longest first, and LIMIT 5.

Part 2: Count by a group

Q3. How many trips happened in each city? Run it and **look carefully**. Does the answer look right? Write a note under the query: what looks wrong?
Hint: use GROUP BY city and COUNT(*).

Part 3: Clean the data

Q4a. Find the **duplicate trips**. A duplicate is the same customer, same driver, same date and same fare, saved two times with different trip_id numbers. Show both trip_id numbers for each one.
Hint: GROUP BY customer_id, driver_id, trip_date, fare, then HAVING COUNT(*) > 1. Use MIN(trip_id) and MAX(trip_id) to see the two numbers.

Q4b. How many **Completed** trips have a **missing fare**?
Hint: for empty values write fare IS NULL. Never write fare = NULL.

Q5. **Fix the data.** (1) Make every city have one correct spelling. (2) Delete the duplicate rows: keep the first one, delete the later copy. (3) Do **not** touch the missing fares. We will report them, not invent numbers. Then **prove it worked**: run your Q3 query again, and count the rows again.
Hint: these are the shapes of the commands:

```
UPDATE table_name SET column = 'new value' WHERE column = 'old value';
UPDATE table_name SET column = TRIM(column);   -- removes extra spaces
DELETE FROM table_name WHERE trip_id IN (1, 2);
```

Part 4: Find the answers (use the clean data)

Q6. **Revenue by City.** For Completed trips only, show each city, the number of trips, the total revenue, and the average fare (rounded to 2 decimals). Biggest revenue first.
Hint: SUM(fare), COUNT(*), ROUND(AVG(fare), 2), GROUP BY city, ORDER BY revenue DESC.

Q7. **Revenue by Month.** For Completed trips only, show each month, the number of trips and the total revenue, in date order. Which month is the best?
Hint: DATE_FORMAT(trip_date, '%Y-%m') turns a date into something like 2026-01.

Part 5: Join two tables

Q8. **Revenue by Vehicle Type.** For Completed trips only, show each vehicle type, the number of trips and the total revenue. Biggest first.
Hint: vehicle_type is in the drivers table, not trips. Join them: FROM trips t INNER JOIN drivers d ON t.driver_id = d.driver_id. Never forget the ON part!

Bonus questions (extra marks)

Q9. Which customers have **never booked** a trip?

Hint: use LEFT JOIN, then look for rows where the trips side is empty (IS NULL).

Q10. Who are the **top 3 customers by total spend** on Completed trips? Show name, total spend and number of trips.

Your message to the manager

Write **3 to 5 simple sentences**. Answer these:

- Which city should ZoomRide invest in? Why? Use numbers from your queries.
- Name **two** data problems you found. What would go wrong if you did not fix them?
- What is one thing you would like to know more about before a big decision?

6. SQL help sheet

A query is written in this order. Read it like a sentence.

```
SELECT column1, column2    -- what to show
FROM table_name            -- which table
WHERE condition            -- keep only some rows
GROUP BY column            -- one result row for each group
HAVING condition           -- filter the groups
ORDER BY column DESC       -- sort (DESC = biggest first)
LIMIT 5;                   -- show only the first 5 rows
```

I want to...	I type...
Count rows	COUNT(*)
Add up numbers	SUM(fare)
Find the average	AVG(fare) or ROUND(AVG(fare), 2)
Find empty values	fare IS NULL
Find values that are not empty	fare IS NOT NULL
Remove extra spaces	TRIM(city)
Get the month from a date	DATE_FORMAT(trip_date, '%Y-%m')
Join two tables	FROM a INNER JOIN b ON a.id = b.id
Keep all rows from the first table	FROM a LEFT JOIN b ON a.id = b.id
Give a column a new name	SUM(fare) AS revenue

Small tips

• Words like 'Completed' go inside single quotes: status = 'Completed'.
• COUNT(*) counts all rows. COUNT(fare) counts only rows where fare is not empty.
• A JOIN without ON mixes every row with every row. The answer will be huge and wrong.
• WHERE filters rows *before* grouping. HAVING filters groups *after* grouping.

Did you learn a different SQL before?

The main ideas of SQL are the same everywhere, but a few commands are different. **For this project, use the MySQL way.**

Task	MySQL (use this)	Other systems
Show only the first 5 rows	LIMIT 5	TOP 5 (SQL Server)
Get the month from a date	DATE_FORMAT(trip_date, '%Y-%m')	strftime (SQLite), TO_CHAR (PostgreSQL, Oracle)
Join pieces of text	CONCAT(a, b)	a || b (PostgreSQL, SQLite, Oracle)

7. What to hand in

- **Your queries.** Copy only what is **below** the line "YOUR QUERIES START HERE". You do not need to send the setup part.
- **Your answer sheet.** The key numbers from Q1 to Q8 (and the bonus if you did it).
- **Your message to the manager.** 3 to 5 sentences.

> **Where do I send it?**
> I will tell you in class where to send your work. Please do not send it before I confirm the place.

How you will be marked

What	Marks
Correct answers (Q1 to Q8)	40
Data cleaning (Q4 and Q5)	20
Neat queries with notes and good names	15
Your message to the manager	25
Bonus (Q9 and Q10)	up to 5 extra

8. If something goes wrong

Problem	What to do
I see an error message	Read the line number in the message. Check for a missing semicolon ; or a spelling mistake. Check that words like 'Completed' have quotes.
It says the table does not exist	You did not run the setup, or you are in a different file. Make sure the setup is pasted at the top of your file, then click Run.
My numbers are different from my friend's	Did you use WHERE status = 'Completed'? Did you run the cleaning (Q5) **before** Q6, Q7 and Q8? Keep your queries in order.
I do not see any answer	Click Run. Then scroll down in the Output box. Each query shows its own table of results.
My file disappeared	Use the same browser and the same device. Keep a backup copy in Notepad or a Google Doc.
I got a huge answer	You may have joined two tables without ON. Add the ON part.
I am stuck	Ask me in class, or ask a classmate. Show your query and the error message.

Good luck! Take it one step at a time. Every expert started as a beginner.